Accelerating AI compute satellite engineering with an AI agent and GPU simulation

Author: Neil Dave assisted by AI agent Grok Bot

Engineers set the requirements. An AI agent runs the analysis. Physics solvers check the selected design.

Human requirements / design decisions → AI agent (closed source) → Open analysis scripts Apache-2.0 → GPU: Warp, PhysicsNeMo, Newton
CPU: CalculiX FEA → Solver checks → Human review / approval

Full repo (GitHub)

PROBLEM

Putting AI data centers in orbit can ease power generation concerns on Earth, but the satellites must be mass- and volume-efficient to meet launch and radiative cooling needs. SpaceX's published targets of 250 kW peak per satellite and 75 kW per ton imply 30 satellites × 250 kW = 7.5 MW installed peak per 100 t Starship launch. An engineer and an AI agent explored the design space and selected a high-level architecture to meet that target.

Source: spacex.com/spacexai/starmind

DESIGN ARCHITECTURE (PER SATELLITE)

Compute: one Vera Rubin NVL72 (72 GPUs + 36 CPUs); 250 kW peak at beginning of life. Assumed 640.38 kg payload allocation in a 600 × 1200 × 937 mm box.

Mass: 3,026 kg per satellite; all 30 satellites fit one 100 t Starship at 90.8 t, leaving 2 t for the dispenser and a 7.2 t reserve.

Power: 862 m² solar array; assumed module rating 324.55 W/m² on its remaining planar area; 175 kW nominal in sunlight and a 164.4 kW annual average, including eclipse.

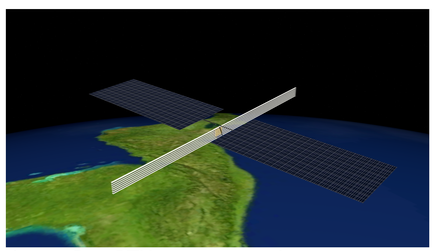
Thermal: Rubin GPU/HBM ≈370 K nominal target (est. 371–373 K), higher in edge cases; junction temperatures not yet qualified. Coolant inlet peaks at 392 K. 140 radiator panels; coating emissivity 0.92. Fins and tubes selected by search; modeled glycol-water loop.

Rack structure: Compute rack screening (CalculiX beam model, 6 g axial / 2 g lateral, FS 1.25): a ≈50 kg aluminum frame inside the 640 kg allocation reaches 35 Hz (Falcon reference) with large strength margins.

Panel vibration: Stowed radiator-panel first mode 128–153 Hz across support-pad assumptions (≈139 Hz nominal), well above the 35 Hz reference.

Folding and deployment: 31/31 representative hinge locks latch in Newton rigid simulation; sampled clearance passes.

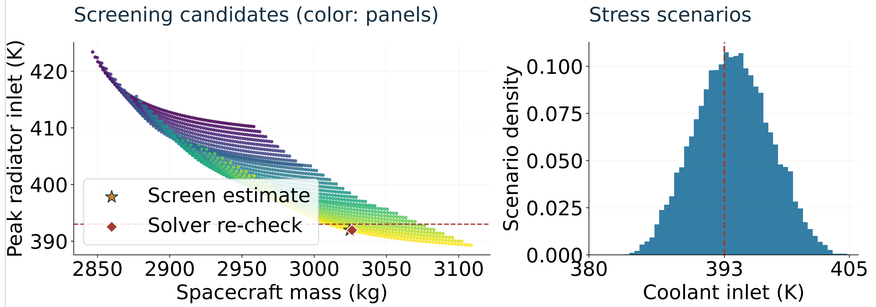
A basis for further detailed design. Simulation study, not flight qualification.



Illustrative Isaac Sim RTX render, not flight validation. Earth: NASA Blue Marble.

1 What did the search find? | Warp

1.81M designs; 3,048 feasible, lightest picked



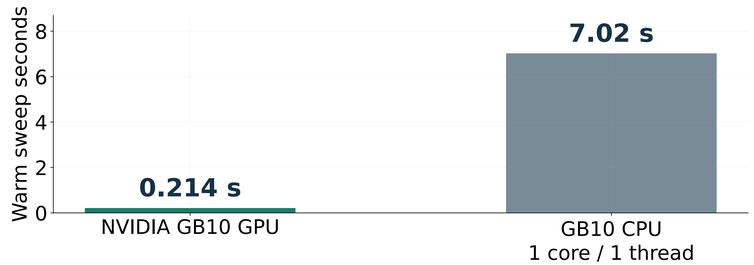
Only 41.66% of assumed scenarios meet the 393 K coolant cap.

1.05M coating / power / sun-exposure scenarios; not flight reliability.

Star: screening estimate. Diamond: solver re-check, 3026.21 kg / 140 panels.

2 What ran faster? | GB10 benchmark

32.9× faster than one GB10 CPU core

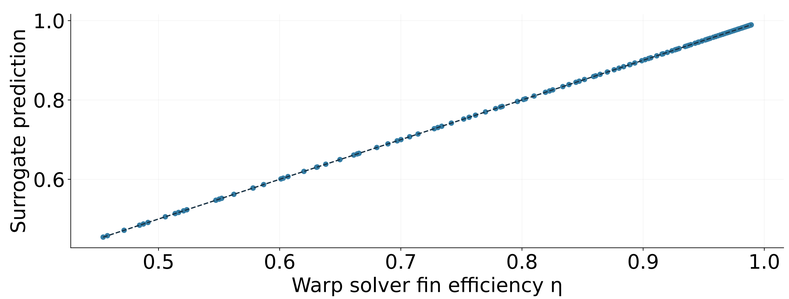


Same fp64 workload; warm kernel only; median of 3 runs. Warp 1.18.0.
1,806,336 combinations; CPU: 1 Cortex-X925 core / 1 thread.
Device-synchronized timer; setup/JIT/readback excluded.

8.46M designs/s; input SHA256 94e1b3ae978b... (full hash in repo).

3 Can we trust the search? | PhysicsNeMo / Tape

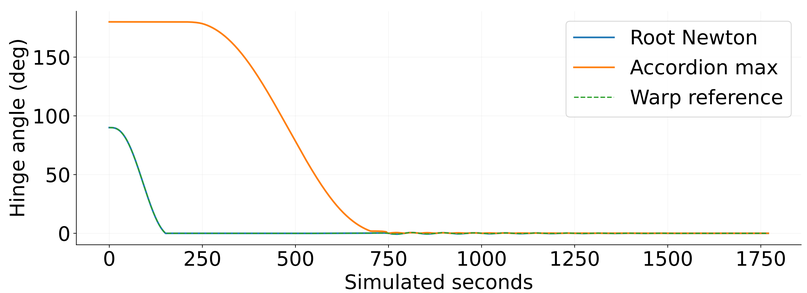
Max |Δη| = 9.3e-5 (held out)



PhysicsNeMo: 384 training / 128 held-out labels; η = fin efficiency.
Selected fins re-solved with Warp; approximation is not flight validation.
warp.Tape: 0.65% relaxed mass gap; gradient check passed.
Continuous relaxation, not adopted hardware or a global optimum.

4 Does it deploy? | Newton / Warp

31 / 31 modeled hinge locks



Rigid panels / fixed bus; representative hinges, not full flight tests.
Independent sampled-clearance check passed; angle parity 0.0061°.
Newton 1.7.0 and Warp 1.18.0 on CUDA; agreement is not hardware validation.
Flexible hardware and force-step convergence remain open.

≈11 days recorded span | ≥44.1 h GB10 computer-job time* | Why it matters: more checks before hardware, not just a faster kernel

*CPU + GPU jobs, including rendering; not GPU-hours. Start: earliest-retained-input timestamp proxy, not verified first problem definition.